FloodCrew: Project Proposal

A smart helper that tells a council where to send its crews and gritters, and warns the public, during floods, snow and black ice.

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1. The problem

When a river floods or roads freeze, a council has only a few crews and gritters. Staff must decide quickly where to send them, often using guesswork or waiting for an official warning, which can be too late. A care home, school or hospital road can be left unprotected. Drivers and people walking also often find a flooded or icy road only when they reach it.

2. Aim

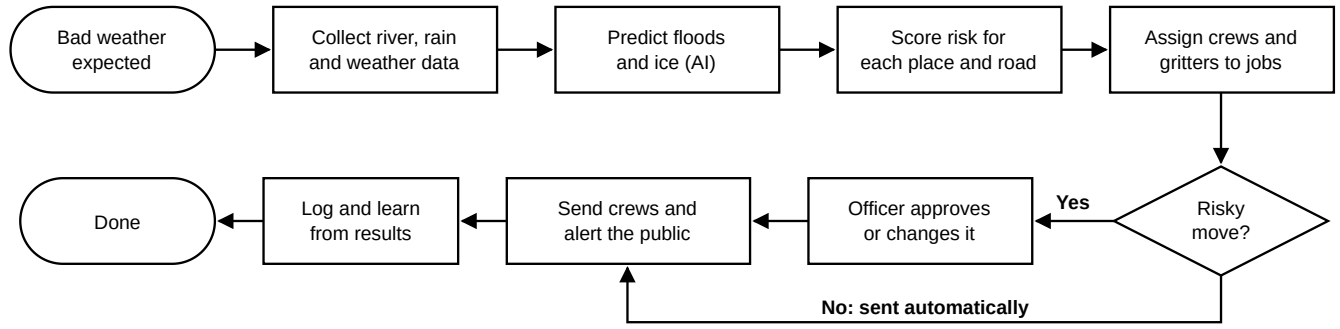
To build a smart system that helps a council **protect the most vulnerable places before bad weather hits**, send its crews and gritters to the right places, and **keep the public informed**.

3. Objectives

1. Predict river levels 6 hours ahead and which roads will freeze overnight.
2. Give every place and road a risk score (how likely × how bad).
3. Suggest the best job for each crew and gritter automatically.
4. Let the officer approve the big decisions and try "what if" scenarios.
5. Send the public alerts for their street and a safer route.
6. Test the system fairly against what councils do today.

4. The proposed solution

How FloodCrew works:



It follows Simon's decision model (intelligence, design, choice, review) and has three parts:

Officer dashboard	**Crew phone page**	**Public alerts page**
Map, ready-made plan, approvals and what-if	Each job and the reason for it, plus status updates	Alerts by postcode, flooded and icy roads on a map, safer route

5. Who will use it

User	What they get
Emergency officer	A ready-made plan, with only the important moves to approve
Crew lead	Jobs on a phone, with the reason for each
Public (drivers, walkers, residents)	Alerts for their street and a safer way to travel

6. Why it is an intelligent decision support system

IDSS part	In FloodCrew
Data	Environment Agency river levels and rainfall; weather forecasts for ice
Models (AI)	Machine learning predicts river levels and road temperature

Knowledge base	Rules: risk = likelihood × impact; extra weight for care homes, schools, hospital routes
Optimiser	Hungarian algorithm gives each crew the best job, counting travel time
User interface	Dashboard, crew page and public alerts page
Human in control	Officer approves risky moves; every action is logged and can be undone

7. Tools and data

- **Data:** Environment Agency open data (free) and Open-Meteo weather forecasts. Simulated River Tees data is used first while real data is collected.
- **Build:** Python (scikit-learn, SciPy, FastAPI) and a web app.

8. How it will be tested

The models are trained on older data and tested on newer data they have never seen. FloodCrew is compared with today's practice: sending crews when the official warning comes, and gritting at fixed times. Early results on simulated data:

Test	FloodCrew	Today's method
River level prediction error	About 3 cm	About 9 cm
Protected sites reached before flooding (5 crews)	12 of 13	7 of 13
Road temperature prediction error	0.85 °C	2.90 °C
Icy roads gritted in time (1 gritter)	79%	50%
Gritting runs used (2 gritters)	412	436

9. Expected benefits

- Vulnerable places protected earlier.
- Better use of a small number of crews and gritters.
- Fewer people driving or walking into danger.
- Clear, explainable decisions with a full record.

10. Risks and limits

- Results so far use simulated data; real data is the next step.
- Flood levels and gritting routes are estimates until the council's real ones are used.
- Public alerts need officer approval to avoid false alarms. The system supports official warnings and does not replace them.